

AP* Chemistry

Atomic Structure & Periodicity

(modified from NMSI resources by René McCormick)

QUICK REVIEW

- Democritus (2000 BC): “Invention” of the concept of the atom
- Dalton (1808): Experimental evidence used to create atomic theory
- Thomson (1897): Discovered electron (first subatomic particle) and proved atoms have internal structure
- Rutherford (1911): Discovered nucleus and placed electrons outside nucleus

THE NATURE OF MATTER & ENERGY

- Max Planck- There is a minimum amount of energy that can be gained/lost by an atom, and all energy gained/lost must be some integer multiple, n , of that minimum.

$$\text{Energy} = (n)h\nu$$

- h is a proportionality constant, Planck's constant, $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$
- ν is the lowest frequency that can be absorbed/emitted
- minimum energy change, $h\nu$, is called a **quantum**.

Frequency and wavelength of light are related:

$$c = \lambda \nu$$

- c is the speed of light, $c = 3.00 \times 10^8 \text{ m/s}$
- wavelength (λ) and frequency (ν) are inversely proportional

THE PHOTOELECTRIC EFFECT AND ALBERT EINSTEIN

Einstein proposed that light (and all EM radiation) was quantized. He proposed that light could be viewed as a stream of “particles” called **photons**.

- **photoelectric effect**: light bombards the surface of a metal and electrons are ejected.
 - Ejection is based upon **frequency** (related to energy by $E = h\nu$) **NOT intensity** (brightness)
 - A minimum ν must be met or nothing happens.
 - Once beyond minimum, higher ν ejects electrons with more energy.
- **photon**: massless particles of light.

$$E_{\text{photon}} = h\nu = \frac{hc}{\lambda}$$

You know Einstein for $E = mc^2$ from his special theory of relativity. This leads to several more equations...

$$m = \frac{E}{c^2} = \frac{hc/\lambda}{c^2} = \frac{h}{\lambda c}$$

EM radiation (light, x-rays, etc.) exhibits wave and particle properties. This is known as the **dual nature of light**.

Since light was thought to be wavelike with characteristics of particulate matter, Louis de Broglie wondered if matter (traditionally modeled as particles) could exhibit wavelike characteristics.

$$m = \frac{h}{\lambda v}$$

This substitutes v (velocity) for c for an object NOT traveling at the speed of light, and rearrange... de Broglie equation:

$$\lambda = \frac{h}{m v}$$

- The more massive the object, the smaller its associated wavelength
- *We now know that energy is really a form of matter, and ALL matter shows the same types of properties.*
ALL matter exhibits both PARTICULATE and WAVE properties (dual nature).

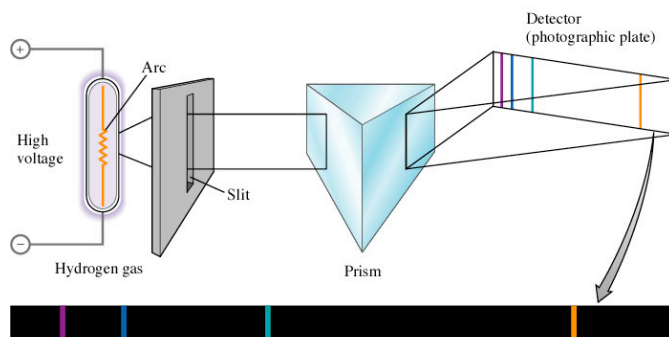
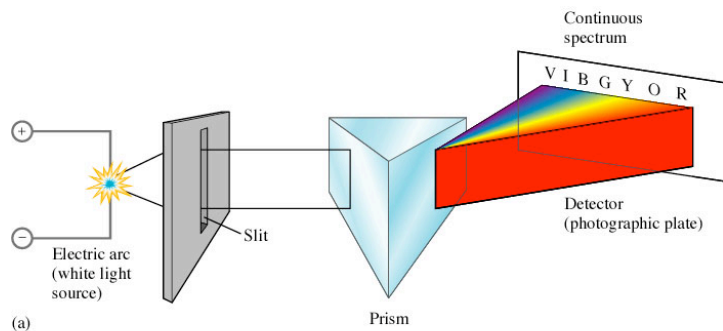
Exercise 1

Compare the wavelength for an electron (mass = 9.11×10^{-31} kg) traveling at a speed of 1.0×10^7 m/s with that for a ball (mass = 0.10 kg) traveling at 35 m/s.

$$\lambda_{\text{electron}} = 7.27 \times 10^{-11} \text{ m}; \lambda_{\text{ball}} = 1.9 \times 10^{-34} \text{ m}$$

HYDROGEN'S ATOMIC LINE SPECTRA AND NIELS BOHR

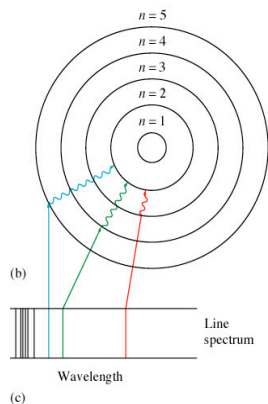
- **emission spectrum:** the spectrum of radiation emitted by a substance as it loses energy
- **absorption spectrum:** the spectrum of radiation absorbed by a substance as it gains energy (as a function of λ)
- **line spectrum:** isolation of a thin beam of emission spectrum by passing through a slit then a prism or a diffraction grating which sorts the radiation into discrete (specific & unique) frequencies or lines



- Niels Bohr connected emission spectra and the quantum ideas of Einstein and Planck:

The electron of the hydrogen atom could occupy only certain energy states.

- Bohr assumed that an electron in an atom would remain in its lowest energy state unless otherwise disturbed.
- Energy is absorbed in order to move the electron from this ground state.
- An electron with $n = 1$ (the ground state) is most strongly attracted to the positive nucleus.
 - Energy absorbed must eventually be emitted.
 - The origin of atomic line spectra is the movement of electrons between **quantized energy states**.
 - If an electron moves from higher energy level to a lower energy state, a photon is emitted and an emission line is observed.
 - Bohr's equation for calculating the energy of the E levels available to the electron in the hydrogen atom:



$$E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$$

- n is an integer [larger n = larger orbit (higher energy level), farther from nucleus]
- Z is the nuclear charge ($Z = 1$ for hydrogen)

- The negative sign means that the E of an electron bound to the nucleus is lower than it would be if the electron were an infinite distance from the nucleus (not "in" the atom) where there is NO interaction and the E is zero.
- ΔE is simply the subtraction of calculating the energy of two different levels, say $n = 6$ and $n = 1$. If the difference is negative, E was lost. If the difference is positive, E was gained.
- Major defects in Bohr's theory:
 - 1) Only works for elements with ONE electron (hydrogen or other atoms after losing e^-).
 - 2) The lone electron does **NOT** orbit the nucleus in a fixed path (had to ignore observations about orbiting bodies that did not support his theory).

Exercise 2

Calculate the energy required to excite the hydrogen electron from level $n = 1$ to level $n = 2$.

$$\Delta E = 1.633 \times 10^{-18} \text{ J}$$

Exercise 3

Calculate the energy required to remove the electron from a hydrogen atom in its ground state.

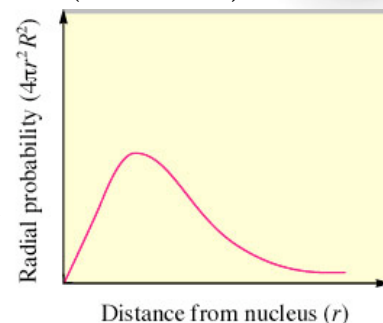
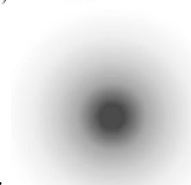
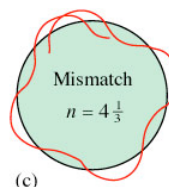
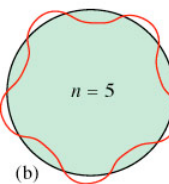
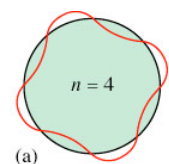
$$\Delta E = 2.178 \times 10^{-18} \text{ J}$$

THE WAVE PROPERTIES OF THE ELECTRON SCHRODINGER, HEISENBERG, AND QUANTUM NUMBERS

- Erwin Schrodinger: focused on de Broglie's equation and the electron as a wave. Schrodinger's approach explained more than Bohr's and met with more success. Quantum mechanics was born.
- Werner Heisenberg and Max Born provided the **uncertainty principle**
 - If you want to define the momentum of an electron, then you have to forego knowledge of its exact position at the time of the measurement
- Max Born on the basis of Heisenberg's work suggested:
 - We can only calculate the **probability** of finding an electron within a given space.

THE WAVE MECHANICAL VIEW OF THE ATOM

- Schrodinger equation: solutions are called wave functions. The electron is characterized as a matter-wave.
- Sort of standing waves: *only certain allowed wave functions* (symbolized by the Greek letter, ψ).
- Each ψ for the electron in the H atom corresponds to an allowed energy. For each integer, n , there is an atomic state characterized by its own ψ and energy E_n .
- Means that the energy of electrons is **quantized**.
- Only whole numbers of standing waves can “fit” in the proposed orbits (diagrams to the right).
 - The hydrogen electron is visualized as a standing wave around the nucleus (above right). The circumference of a particular circular orbit would have to correspond to a whole number of wavelengths, as shown in (a) and (b) above, OR else destructive interference occurs, as shown in (c). This is consistent with the fact that only certain electron energies are allowed; the atom is quantized. (*Although this idea encouraged scientists to use a wave theory, it does not mean that the electron really travels in circular orbits.*)
- Matter-waves for allowed energy states are also called orbitals.
- Electron density map, electron density, and electron probability all mean the same thing. When we say orbital a “fuzzy cloud” is what we picture in our minds (to the right).
- To solve Schrodinger's equation in a 3-dimensional world we use the quantum numbers (n , ℓ , and m_ℓ).
- Plot the total probability of finding the electron in each shell versus the distance from the nucleus and you get the radial probability graph you see in to the right.
- The maximum in the curve occurs because of two opposing effects.
 1. the probability of finding an electron is greatest near the nucleus, BUT
 2. the volume of the spherical shell increases with distance from the nucleus, SO we are summing more positions of possibility, so the TOTAL probability increases to a certain radius and then decreases



QUANTUM NUMBERS & ATOMIC ORBITALS

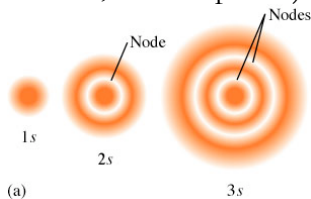
- **n : principal quantum number**- values range from 1 to infinity. Indicates the probable distance (related to energy) of the electron from the nucleus (measure of orbital size). Electron's **ENERGY LEVEL**.
- **ℓ : angular momentum quantum number**- values range from 0, 1, 2, ...($n-1$). Electrons within level/shell are grouped into subshells, characterized by its wave shape (s, p, d, f). Each ℓ is a different orbital shape/type and corresponds to the number of nodal planes.
- **m_ℓ : magnetic quantum number**- describes the orientation (x/y/z-axis) of an orbital in a given subshell; values = integers from $-\ell$ to $+\ell$.
- **m_s : magnetic spin quantum number**- accounts for attraction to magnetic field; values = $+\frac{1}{2}$ or $-\frac{1}{2}$

THE SHAPES OF ATOMIC ORBITALS (different ℓ 's)

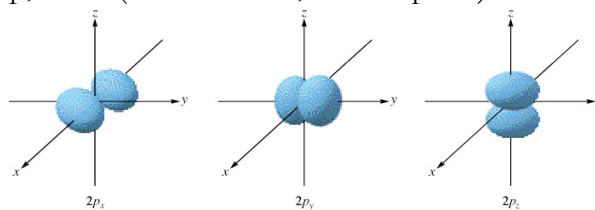
There is no sharp boundary beyond which the electrons are never found!!

- Nodal plane- the electron can never be found there!!

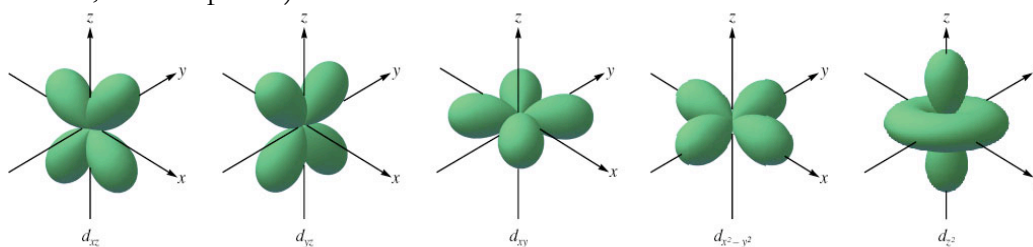
s, $\ell = 0$ (1 orientation, 0 nodal planes)



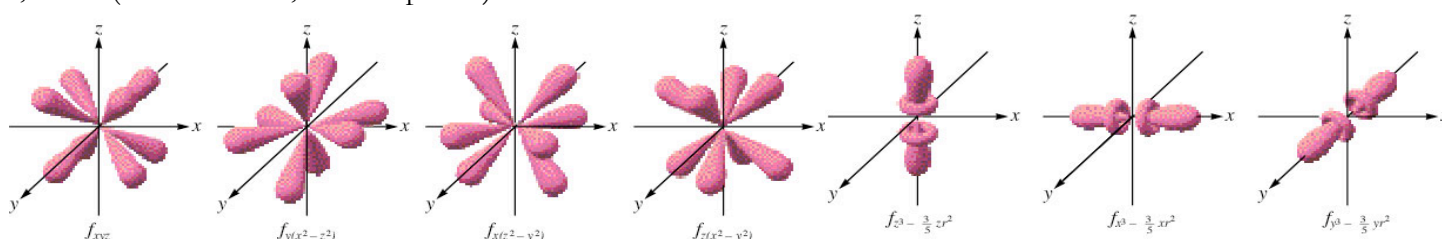
p, $\ell = 1$ (3 orientations, 1 nodal plane)



d, $\ell = 2$ (5 orientations, 2 nodal planes)



f, $\ell = 3$ (7 orientations, 3 nodal planes)



ELECTRON SPIN

- Electrons interact with a magnetic field, which is explained by describing electron spin.
- **diamagnetic**- not magnetic (magnetism dies); in fact they are *slightly* repelled by a magnetic field.
 - Occurs when all electrons within atom are **PAIRED**.
- **paramagnetic**- attracted to a magnetic field; lose their magnetism when removed from the magnetic field
 - Atom has one or more **UNPAIRED** electrons.
- **These are explained by electron spins**
 - Electron spin is quantized such that only two orientations of the electron and its spin are possible.
 - Called $+\frac{1}{2}$ and $-\frac{1}{2}$.
 - Cancel when paired (gives evidence for Pauli Exclusion Principle).
- **ferromagnetic**- retain magnetism upon introduction to, then removal from a magnetic field
 - Electron spins from one atom influence spins in neighboring atoms.
 - Most stable arrangement is when spins on adjacent atoms are aligned; causes solid to behave as a magnet.
 - If dropped, the domains can unalign and it no longer operates as a magnet.

RULES FOR ELECTRON CONFIGURATIONS

- Pauli Exclusion Principle: no two electrons in an atom can have the same set of four quantum numbers
 - No orbital can contain more than 2 electrons & must have opposite spins in the same orbital
 - s = 1 orbital, 2 e⁻ p = 3 orbitals, 6 e⁻ d = 5 orbitals, 10 e⁻ f = 7 orbitals, 14 e⁻
- Aufbau Principle: electrons fill lowest-energy orbitals completely before filling higher-energy orbitals
 - Order of orbital-energy is based upon organization of the periodic table
 - 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p, 8s...
 - Weird order (4s before 3d, etc.) comes from how electrons interact as orbitals are filled
- Hund's Rule: The most stable arrangement of electrons has the maximum number of unpaired electrons
 - Will place 1 electron in each orbital within a sublevel before doubling up in any orbital
 - This minimizes electron-electron repulsions

Configuration Examples (*hopefully, review!!!*)

S 1s² 2s² 2p⁶ 3s² 3p⁴

Cd 1s² 2s² 2p⁶ 3s² 3p⁶ 4s² 3d¹⁰ 4p⁶ 5s² 4d¹⁰

La [Kr] 5s² 4d¹⁰ 5p⁶ 6s² 5d¹

Hf [Xe] 6s² 4f⁴ 5d²

Ra [Rn] 7s²

Ac [Rn] 7s² 6d¹

	1A	2A																3A	4A	5A	6A	7A	8A
1	1s																						1s
2	2s																			2p			
3	3s																		3p	S			
4	4s						3d												4p				
5	5s						4d												5p				
6	6s		La	Hf			5d												6p				
7	7s	Ra	Ac				6d																
																		4f					
																		5f					

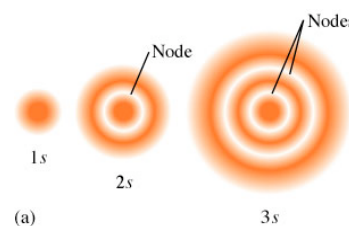
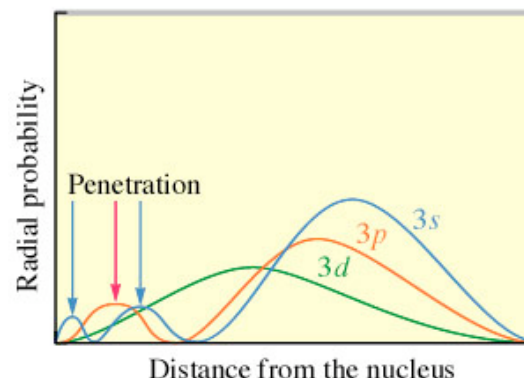
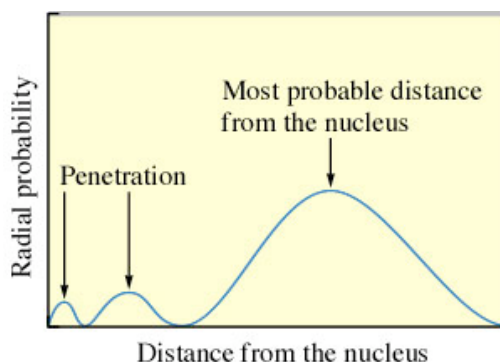
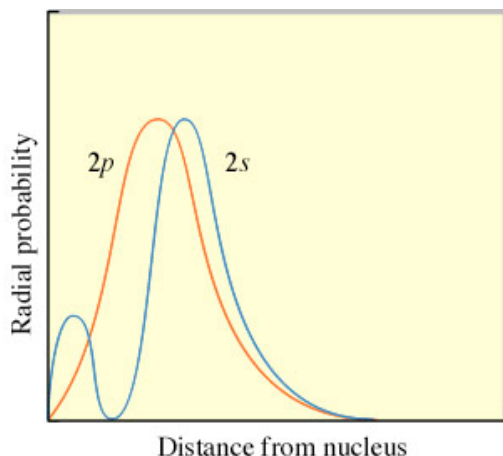
EXPLANATIONS FOR THE AP EXAM (rare, but important)

Fundamental scientific principles for explanations of electron energies:

- electrons repel each other
- electrons are negative and attracted by the positive nucleus
- forces (attraction/repulsion) dissipate with increasing distance

Graphs

- Tallest peak is most probably distance for electron
- Farther left is closer to nucleus (electron penetrating closer to nucleus)
 - Look at location of tallest peak
 - Look at location of closest peak
 - Least penetrating are easiest to remove



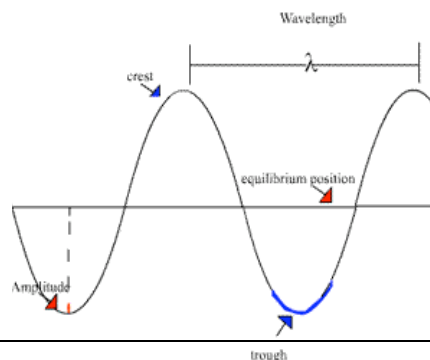
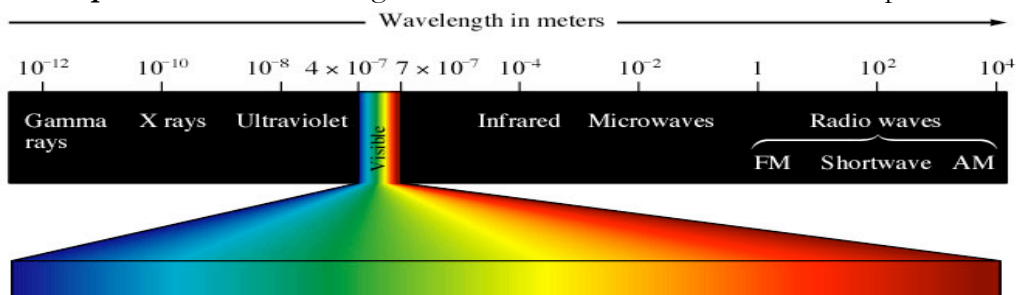
ION ORBITAL ENERGIES AND ELECTRON CONFIGURATIONS

- Atoms lose the s-electrons from the highest level (largest *n*) before losing d/f electrons.
- Many transition metals have +2 charge on cation
- As electrons are removed, orbital energies "spread out" so the difference between s, d, and f orbitals becomes greater (weird filling order of some transition metals is eliminated)

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ELECTRONS & ELECTROMAGNETIC RADIATION

- **electromagnetic radiation:** UV, visible light, IR, microwaves, gamma, radio waves, and X-rays
- **wavelength (λ):** (lambda) distance between crests.
- **frequency (ν):** (nu in chemistry; f in physics), number of *cycles per second* that pass a certain point in space
- **amplitude:** maximum height of a wave as measured from the “midpoint”



Exercise 4

The blue color in fireworks is often achieved by heating copper compounds. At high temperatures, the compounds emit a blue light with a wavelength of 450 nm. What amount of energy is associated with this light?

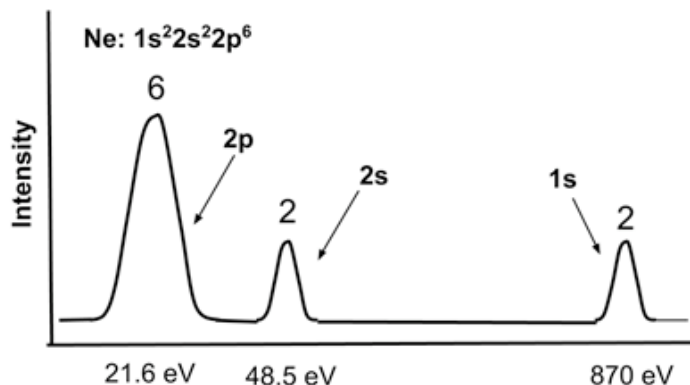
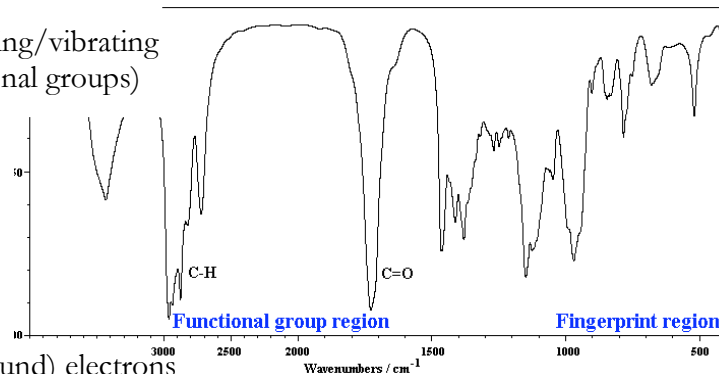
$$E = 4.41 \times 10^{-19} \text{ J}$$

SPECTROSCOPY

Just like Bohr used hydrogen's visible spectrum to explore the structure of the atom, other forms of electromagnetic radiation can help probe the structure of matter through a process called **spectroscopy**.

(Listed lowest to highest energy... least to most effect on substance)

- **NMR (nuclear magnetic resonance):** change in nuclear spin caused by **radio waves** used to reveal bonding patterns, especially in organic molecules
- **Infrared spectroscopy:** bond stretching/compressing/bending/vibrating caused by IR photons that identifies specific bonds (functional groups)
- **Visible & UV spectroscopy:** movement of valence electrons from ground state to excited state(s)
 - Gives some transition metal solutions color
 - Relates to Bohr's line spectra
 - Can sometimes remove the valence electron(s) (photoionization/photoelectric effect)
- **X-rays & PES:** removal of core electrons by X-rays
 - Uses lots of energy to remove inner (more tightly bound) electrons
 - PES spectra indicate number of energy levels, sublevels, and electrons
 - **Must be able to interpret (read), draw, and compare PES graphs**



THE HISTORY OF THE PERIODIC TABLE

1800ish- Johann Dobereiner, triads

1864- John Newlands, octaves

1870- Dmitrii Mendeleev & Julius Lothar Meyer, by mass

1913- Mosley, by number of protons (periodic law)

- Group 1- Alkali metals
- Group 2- Alkaline earth metals
- Groups 3-12- Transition metals
- Group 15- Pnictogens
- Group 16- Chalcogens
- Group 17- Halogens
- Group 18- Noble gases

SOME PROPERTIES OF COMMON GROUPS:

- Alkali metals- the most reactive metal family; react with violently water
- Alkaline earth metals- metal hydroxides formed by this group provide basic solutions in water
- Halogen family- the most reactive nonmetal family
- Noble gas family- known for their chemical disinterest in other elements; once thought to never react
- Transition metals- fill the d-orbitals
 - Anomalies in the aufbau principle occur at Cr and Cu (& others) to *minimize electron/electron repulsions*.
 - There is NO special stability associated with half-filled sub-levels.

K 4s ¹	Ca 4s ²	Sc 3d ¹	Ti 3d ²	V 3d ³	Cr 4s ¹ 3d ⁵	Mn 3d ⁵	Fe 3d ⁶	Co 3d ⁷	Ni 3d ⁸	Cu 4s ¹ 3d ¹⁰	Zn 3d ¹⁰	Ga 4p ¹	Ge 4p ²	As 4p ³	Se 4p ⁴	Br 4p ⁵	Kr 4p ⁶
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- Rare Earth metals- fill the f sublevels.
 - Lanthanides and Actinides. These sometimes put an electron in *d* (just one or two electrons) before filling *f* (known as “dsf overlay”) because the energies of the sublevels are very similar.

PERIODIC TRENDS

This is **THE** important section- there is often free response questions and very wordy multiple choice involving this topic on the AP exam. There are several arguments you will evoke to **EXPLAIN** a periodic trend. Remember opposites attract and likes repel. The trick is learning which argument to use when explaining a certain trend!

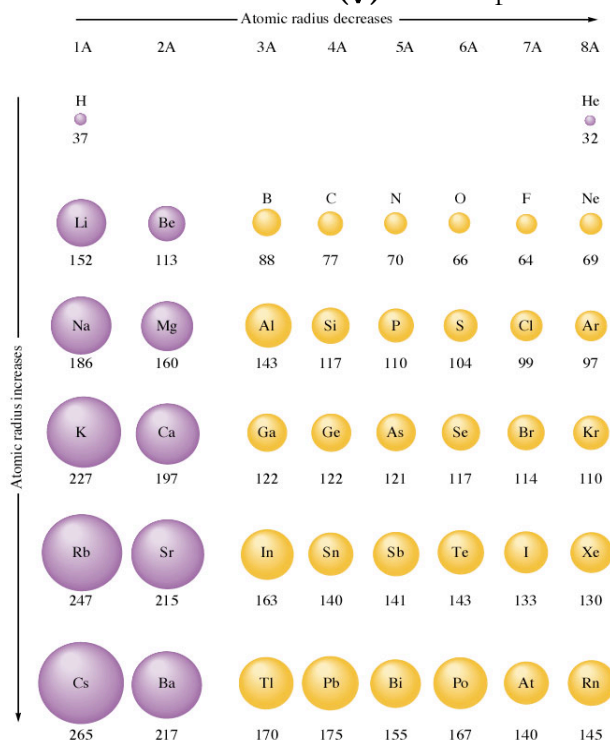
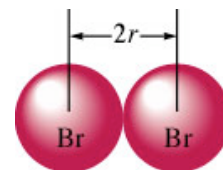
THE ARGUMENTS:

- Effective nuclear charge, Z_{eff} (# of p^+ – core electrons)- essentially equal to the group number.
Group 1: $Z_{\text{eff}} = 1$, Group 3: $Z_{\text{eff}} = 3$, Group 7: $Z_{\text{eff}} = 7$
The idea is that the higher the Z_{eff} , the more positive the nucleus, the more attractive force there is drawing electrons in or holding them in place.
- Distance- attractive forces dissipate with increased distance. Distant electrons (higher energy level) experience less inward pull from the nucleus and are more easily removed.
- Shielding- electrons in the core (non-valence levels) effectively “block” the nucleus’s attractive force for the valence electrons. Use this ONLY when going up and down the table, NOT across. Sublevels do not effectively shield nuclear charge (i.e. 2s sublevel does NOT shield 2p).
- Minimize electron/electron repulsions- this puts the atom at a lower energy state and makes it more stable.

Relate arguments to ENERGY and FORCE OF ATTRACTION whenever possible

1. ATOMIC RADIUS- distance from nucleus to “farthest” electron

- Use diatomic molecules to determine radius
- ATOMIC radii decreases (↓) across a period AND increases (↑) down a column (family)



- **WHY ↓ across?** The effective nuclear charge (Z_{eff}) increases (more protons for the same number of energy levels) as we move from left to right across the periodic table, so the electron experience a greater positive charge from the nucleus, thus the entire electron cloud is more strongly attracted and “shrinks” it.
 - Electron cloud shrinks *until...* the point at which electron/electron repulsions overcome the nuclear attraction and stop the contraction of the electron cloud.
- **WHY ↑ down?** The principal level, n , determines the size of an atom- add another energy level and the atoms get MUCH larger radii.
- As we move down a family, the attractive force the nucleus exerts on the valence electrons decreases.
- Shielding is only a valid argument when comparing elements from one row to another (up and down the table) since shielding within a period is essentially the same.

2. IONIZATION ENERGY- energy required to remove an electron from the atom in the gas phase.

Always **REQUIRES** Energy!!! (even for alkali metals... they do **NOT** “want: to lose”)

- Ionization Energy decreases (↓) down a column (family) AND increases (↑) across a period

- **Why ↓ down?** increased distance from the nucleus and increased shielding by full energy levels means it requires less energy to remove an electron
- **Why ↑ across?** effective nuclear charge (Z_{eff}) increases, so the electron experience a greater positive charge from the nucleus, thus attracting the valence electrons more strongly, more energy required to remove a valence electron

- Removing each subsequent electron requires more energy:
 - First IE < Second IE < Third IE, etc. (*ALWAYS*)

- After removing an electron, the remaining electrons “feel” more attraction from the remaining protons
 - It is harder to remove the second electron from Mg than it is the first
- A **HUGE** energy price is required if the subsequent removal of electrons is from another sublevel or a lower energy level (core)
 - The third electron is VERY hard to remove from Mg (because all valence have been removed)
 - “Jump” in ionization energy is smaller when transitioning from s to p orbitals than from valence electrons to core electrons
- Noble gases aren’t stable just because they have full energy levels, they are unreactive because it takes so much energy to remove electrons

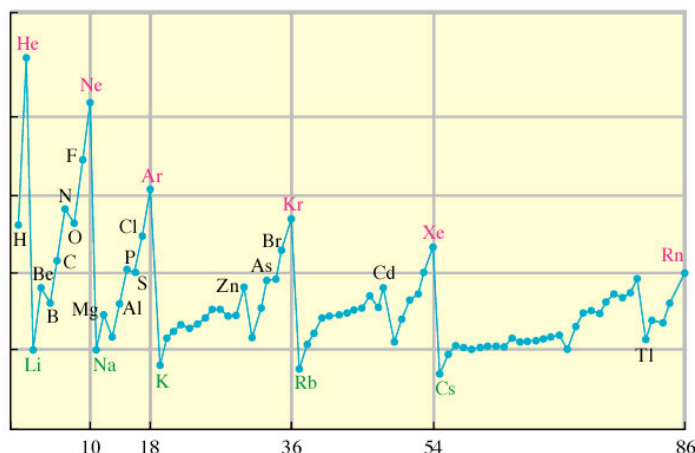
H	
1311	
Li	Be
520	899
Na	Mg
495	735
K	Ca
419	590
Rb	Sr
409	549
Cs	Ba
382	503

					He
					2377
B	C	N	O	F	Ne
800	1086	1402	1314	1681	2088
Al	Si	P	S	Cl	Ar
580	780	1060	1005	1255	1527
Ga	Ge	As	Se	Br	Kr
579	761	947	941	1143	1356
In	Sn	Sb	Te	I	Xe
558	708	834	869	1009	1176
Tl	Pb	Bi	Po	At	Rn
589	715	703	813	(926)	1042

EXCEPTIONS:

- An anomaly occurs at a half-filled or filled sublevel. When electron pairing first occurs within an orbital, there is an increase in electron/electron repulsions which makes it require less energy (easier) to remove an electron thus the ionization energy drops. **NEVER write about the fictitious stability of a $1/2$ -filled shell.**

- Oxygen vs. Nitrogen: it requires less energy to remove an electron from oxygen's valence despite larger Z_{eff} because oxygen's p^4 electron is the first to pair within the orbital thus experiencing increased e/e repulsion. The increased e/e repulsion lowers the amount of energy required to remove the newly paired electron.
- Also a drop in IE from any s^2 to p^1 (despite increased Z_{eff}). This drop in the energy required to remove a valence electron is due to the fact that you are removing a p electron rather than an s electron. The p 's are less tightly held because they do not penetrate toward the nucleus as well as an s electron. (They aren't "farther")



3. ELECTRON AFFINITY- an affinity or "liking" for electrons

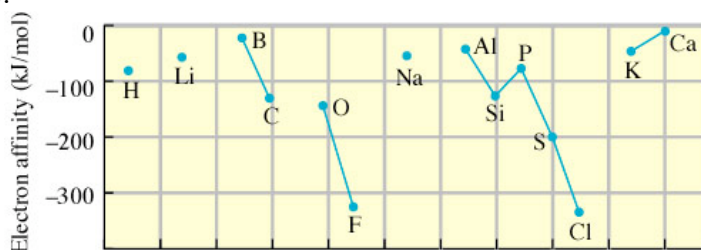
- Energy associated with the addition of an electron to a gaseous atom: $X(g) + e^- \rightarrow X^-(g)$

IF the addition of an electron is exothermic (releasing energy... thermodynamically favorable), you'll see a negative sign on the energy change (positive for endothermic). The more negative the quantity, the more energy is released.

- ↓ down a family (becomes less negative, smaller absolute value... giving off less energy) due to increased distance from the nucleus with each increasing energy level. The nucleus is farther from the valence level and more shielded. Less attractive means less energy is released when electron is gained.
- ↑ across a period (becomes more negative, larger absolute value... releasing more energy). Larger Z_{eff} more strongly attracts the electron; more energy released. *Electron/electron repulsions cause lots of exceptions across a period.*

EXCEPTIONS:

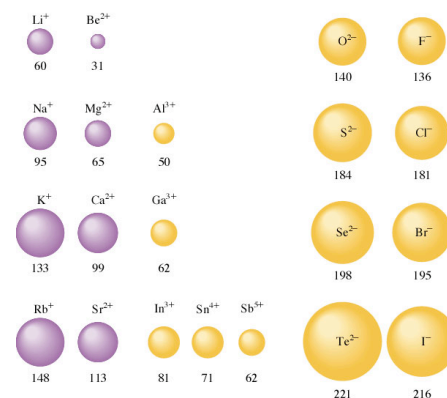
The lines on the diagram below connect adjacent elements. The absence of a line indicates missing elements whose atoms do not add an electron exothermically and thus do not form stable isolated anions (remember these are all forming -1 ions at this point).



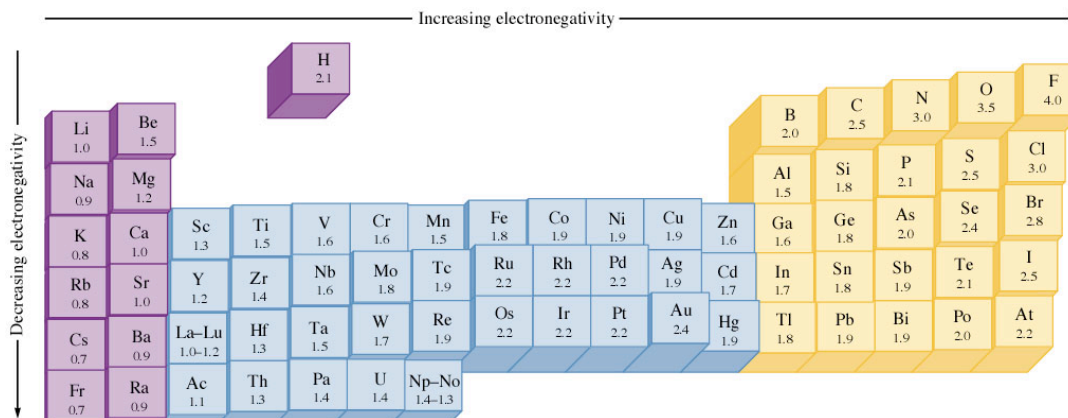
- No N^- yet there is a C^- : due to changes in electron/electron repulsions explained by their electron configurations. N is p^3 while C is p^2 . C adds an electron without pairing which does not increase the e/e repulsions and therefore, carbon forms a stable -1 ion while N would have to pair p electrons and the increased e/e repulsions overcome the increasing attractive force due to the increase in Z_{eff} and no -1 N ion forms.
- O^{2-} doesn't exist in isolated form for the same reason. It's p^4 , so adding the first electron causes a subsequent pairing but it has a greater Z_{eff} than N, so it can form O^- . Adding the second electron fills the p 's and that increased e/e repulsion overpowers the Z_{eff} of oxygen. (Does exist in lots compounds)
- F is weird, it has really strong e/e repulsion since the p orbitals are really small in the second level therefore repulsions are high. In subsequent halogen orbitals, it's not as noticeable. (F is an "exception" for its family)

4. IONIC RADII

- **Cations:** shrink because the nucleus is now attracting fewer electrons
- **Anions:** expand because the nucleus is now attracting more electrons than there are protons and there is enhanced electron/electron repulsion
- **Isoelectronic:** ions containing the same number of electrons
 - Consider the # of protons to determine size: oxide vs. fluoride. Fluoride has one more proton which further attracts the electron cloud, so it is smaller.



5. ELECTRONEGATIVITY- The ability of an atom IN A MOLECULE to attract shared electrons to itself.



- Why is F the most? Highest Z_{eff} and smallest so that the nucleus is closest to the valence “action”
- Why is Fr the least? Lowest Z_{eff} and largest so that the nucleus is farthest from the “action”
- We’ll use this concept a great deal in our discussions about **BONDING** since this atomic trend is only used when atoms form molecules.

Exercise 5

Why is the ionization energy of sulfur less than that of phosphorus?

Identify the atoms from the following electron configurations:

- $1s^2 2s^2 2p^6$
- $1s^2 2s^2 2p^6 3s^1$
- $1s^2 2s^2 2p^6 3s^2$

Which atom has the largest first ionization energy? Which one has the smallest second ionization energy?

Exercise 6

Predict and explain the trend in radius for the following ions: Be^{2+} , Mg^{2+} , Ca^{2+} , and Sr^{2+} .

Exercise 7

Explain why sodium has a negative value for electron affinity but magnesium and neon have positive values.

Exercise 8

Rank the following metals from smallest to largest radius: Al, Ca, Na. Explain.